Explainable AI Clinical Decision Support System

Clinical Prediction Report

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Patient Information

Patient ID	PAT004
Patient Name	Sarah Wilson
Doctor	Dr. Vinoth
Hospital	AIIMS Hospital
Age	56
Gender	Female
Chest Pain Type	Non-anginal Pain
Resting Blood Pressure	138 mmHg
Cholesterol	245 mg/dL
Fasting Blood Sugar	No
Rest ECG	ST-T Wave Abnormality
Maximum Heart Rate	142
Exercise Angina	No
ST Depression	1.4
ST Segment Slope	Flat
Major Vessels	1
Thalassemia	Fixed Defect

Prediction Result

Prediction	Heart Disease
Probability	53.78 %
Risk Level	Moderate

Clinical Recommendation

There are signs associated with heart disease. Consult a cardiologist and undergo additional clinical evaluation.

Top Influencing Features (SHAP)

Feature	Impact
Sex	-0.9289
Thalassemia	0.5196
Major Vessels	0.3648
Exercise Induced Angina	-0.2864
Slope of ST Segment	0.2661

LIME Explanation

Feature	Impact
cp <= -0.16	-0.1561
-0.88 < thal <= 1.18	0.0954
exang <= -0.70	-0.0845
fbs <= -0.41	0.0741
sex <= -1.46	-0.0623

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • Flask • React

Doctor Signature: _____

Patient Signature: _____